

Appendix: Exclusions

Population	Source	Retained	Excluded	Excluded (%)
Proteins: RefDB roster to final static OpenFold3 cohort	2,382	2,192	190	7.98%
Atom assignments within accepted protein mappings	1,996,670	1,991,084	5,586	0.28%

Table 1–Exclusions: Protein and atom-assignment retention.

Stage	Input	Excluded	Retained	Removal basis
RefDB-to-PDB lookup	2,382	22	2,360	No curated primary PDB found by lookup
OpenFold3 topology build	2,360	3	2,357	No usable topology produced
Feature extraction	2,357	11	2,346	No accepted record produced
Geometry screening	2,346	26	2,320	Excluded by automated screen
Experimental target mapping	2,320	69	2,251	Mapper retained too few unambiguous targets
Static model feature check	2,251	42	2,209	Required MOPAC features absent from accepted extraction
Final whole-protein screen	2,209	17	2,192	16 carbonyl-axis flags; 1 identifier-screen exclusion
Total	2,382	190	2,192	7.98% of RefDB source roster

Table 2–Exclusions: Protein retention through the staged pipeline.

Excluded proteins

Stage	Why removed	n	RefDB/BMRB entries
RefDB-to-PDB lookup	Lookup found no curated primary PDB	22	bmr237, bmr1474, bmr2197, bmr2261, bmr4020, bmr4111, bmr4150, bmr4595, bmr4651, bmr4652, bmr4653, bmr4655, bmr5074, bmr5517, bmr5539, bmr5815, bmr5901, bmr5956, bmr6069, bmr15408, bmr16340, bmr17147
OpenFold3 topology build	Topology builder produced no usable topology	3	bmr4979, bmr5348, bmr5519
Feature extraction	Extractor produced no accepted record	11	bmr4040, bmr4174, bmr4402, bmr5503, bmr6515, bmr6715, bmr6755, bmr7305, bmr7306, bmr15277, bmr15278
Geometry screening	Automated geometry-QC screen excluded the record	26	bmr376, bmr1128, bmr1379, bmr2278, bmr4112, bmr4384, bmr4387, bmr4966, bmr5199, bmr5315, bmr5389, bmr5514, bmr5859, bmr6024, bmr6049, bmr6057, bmr6221, bmr6337, bmr6432, bmr6645, bmr6809, bmr6849, bmr15329, bmr16605, bmr16940, bmr17064
Experimental target mapping	Mapper did not retain enough unambiguous targets	69	bmr262, bmr263, bmr264, bmr397, bmr507, bmr4068, bmr4101, bmr4130, bmr4265, bmr4272, bmr4281, bmr4339, bmr4373, bmr4391, bmr4577, bmr4596, bmr4597, bmr4598, bmr4701, bmr4791, bmr4875, bmr4879, bmr4914, bmr4936, bmr4974, bmr5014, bmr5159, bmr5231, bmr5303, bmr5404, bmr5483, bmr5659, bmr5686, bmr5766, bmr5850, bmr5911, bmr5931, bmr5939, bmr6117, bmr6138, bmr6187, bmr6228, bmr6341, bmr6396, bmr6453, bmr6454, bmr6601, bmr6730, bmr6758, bmr6826, bmr6906, bmr7120, bmr7231, bmr7250, bmr10012, bmr15290, bmr15433, bmr15434, bmr15517, bmr15813, bmr16060, bmr16121, bmr16327, bmr16555, bmr16669, bmr16803, bmr17145, bmr17382, bmr17392

continued

Stage	Why removed	n	RefDB/BMRB entries
Static model feature check	Executed model required MOPAC features absent from the accepted extraction	42	bmr5, bmr248, bmr256, bmr284, bmr1129, bmr3485, bmr4217, bmr4396, bmr4446, bmr4451, bmr4461, bmr4684, bmr4685, bmr4755, bmr4859, bmr4897, bmr5024, bmr5240, bmr5241, bmr5459, bmr5470, bmr5860, bmr6128, bmr6290, bmr6326, bmr6329, bmr6335, bmr6442, bmr6709, bmr7236, bmr7262, bmr10110, bmr10322, bmr11086, bmr11358, bmr11359, bmr11360, bmr11361, bmr11362, bmr11371, bmr15280, bmr16328
Final whole-protein screen	Carbonyl-axis screen flagged the record	16	bmr4321, bmr4775, bmr4792, bmr5650, bmr5798, bmr5879, bmr5892, bmr5946, bmr6015, bmr6058, bmr6134, bmr6195, bmr6342, bmr6679, bmr7184, bmr15166
Final whole-protein screen	Pre-model identifier screen excluded the record	1	bmr5867

Table 3–Exclusions: Excluded protein identifiers and removal basis.

Omitted atom assignments

Why omitted	Assignments	Mapped sample (%)
Mapper could not establish both sites for an unresolved assignment pair	1,772	0.0887%
Mapper could not construct a complete alternative assignment	2,149	0.1076%
Mapper placed multiple source assignments on one prepared atom	657	0.0329%
Mapper found no retained BMRB match for a source assignment	337	0.0169%
Correction check did not reconcile RefDB and retained BMRB values	148	0.0074%
Parser rejected an ambiguity or uncertainty field	53	0.0027%
Prepared-structure lookup did not find the named atom	28	0.0014%
Mapper could not pair duplicate assignment identities uniquely	24	0.0012%
Mapper could not construct a consistent pseudoatom image	8	0.0004%
Mapper found no unique N-terminal H target among H1, H2 and H3	244	0.0122%
Prepared HIE state did not contain the named HIS HD1	161	0.0081%
Deprotonated ASP state did not contain the named HD2	3	0.0002%
Disulfide CYS state did not contain the named HG	2	0.0001%

Table 4–Exclusions: Omitted atom assignments by mapping or preparation outcome.

Nucleus	Omitted assignments	Mapped sample (%)
H	3,981	0.1994%
C	1,433	0.0718%
N	172	0.0086%

Table 5–Exclusions: Omitted atom assignments by nucleus.

Proteins with omitted atom assignments

Counts are omitted assignment rows in each prepared population. This roster is taken at mapping acceptance, so a protein can also appear in the exclusion table if it was removed at a later feature-availability or whole-protein screen. A dash indicates that the protein is absent from that population or has no omissions in it.

Protein	Static	MD	Protein	Static	MD	Protein	Static	MD	Protein	Static	MD
bmr49	1	–	bmr4712	25	–	bmr5843	1	–	bmr6738	14	–
bmr247	1	–	bmr4714	2	–	bmr5844	1	–	bmr6739	23	–
bmr248	1	–	bmr4716	19	–	bmr5846	1	–	bmr6746	1	–
bmr256	1	–	bmr4727	29	–	bmr5860	33	–	bmr6747	1	–
bmr260	2	–	bmr4739	10	–	bmr5866	2	–	bmr6754	1	–
bmr274	2	–	bmr4740	4	–	bmr5867	12	12	bmr6759	1	–
bmr294	6	–	bmr4755	2	–	bmr5873	11	11	bmr6761	1	–
bmr295	6	–	bmr4760	4	–	bmr5875	2	2	bmr6769	1	–
bmr326	1	–	bmr4771	27	–	bmr5879	1	1	bmr6771	1	–
bmr443	2	–	bmr4775	1	–	bmr5880	12	–	bmr6776	5	–
bmr555	2	–	bmr4782	2	–	bmr5884	24	–	bmr6779	1	–
bmr932	1	–	bmr4786	34	–	bmr5891	19	–	bmr6782	1	–
bmr979	8	–	bmr4789	16	–	bmr5892	10	–	bmr6787	4	–
bmr1061	4	–	bmr4798	1	–	bmr5898	3	–	bmr6807	1	–
bmr1062	4	–	bmr4814	6	–	bmr5902	4	–	bmr6820	3	–
bmr1162	1	–	bmr4819	1	–	bmr5905	1	–	bmr6821	3	–
bmr1374	1	–	bmr4829	2	–	bmr5908	1	–	bmr6824	2	–
bmr1517	2	–	bmr4834	2	–	bmr5909	1	–	bmr6843	18	–
bmr1658	2	–	bmr4836	4	–	bmr5910	1	–	bmr6850	11	–
bmr1673	2	–	bmr4839	27	–	bmr5912	4	–	bmr6851	2	–
bmr1918	1	–	bmr4840	3	–	bmr5924	2	–	bmr6866	5	–
bmr2060	4	–	bmr4852	14	–	bmr5933	16	–	bmr6872	1	–
bmr2065	2	–	bmr4857	5	–	bmr5935	2	–	bmr6882	38	–
bmr2281	1	–	bmr4858	2	–	bmr5940	8	–	bmr6893	2	–
bmr2410	4	–	bmr4859	1	–	bmr5946	24	–	bmr6894	2	–
bmr2868	1	–	bmr4863	1	1	bmr5948	1	–	bmr6899	20	–
bmr3441	1	–	bmr4868	28	–	bmr5949	1	–	bmr6902	21	–
bmr3442	2	–	bmr4874	1	–	bmr5952	5	–	bmr6913	2	–
bmr4012	46	–	bmr4877	44	–	bmr5958	1	–	bmr6918	6	–
bmr4022	1	–	bmr4878	1	–	bmr5961	2	–	bmr6920	1	–
bmr4033	1	–	bmr4882	6	–	bmr5963	1	1	bmr6928	19	–
bmr4035	8	–	bmr4884	6	–	bmr5968	3	–	bmr6932	5	5
bmr4036	6	–	bmr4886	17	–	bmr5974	2	–	bmr6953	6	–
bmr4037	3	–	bmr4893	2	–	bmr5998	25	–	bmr6969	6	–
bmr4039	4	–	bmr4896	1	–	bmr5999	3	–	bmr6970	1	–
bmr4041	5	–	bmr4897	1	–	bmr6002	22	–	bmr6973	1	–
bmr4043	6	–	bmr4899	2	–	bmr6012	24	–	bmr6980	30	–
bmr4044	1	–	bmr4900	2	–	bmr6015	3	3	bmr6981	11	11
bmr4045	20	–	bmr4908	36	–	bmr6022	7	–	bmr6989	1	1
bmr4047	12	–	bmr4909	24	–	bmr6023	2	–	bmr7020	2	–
bmr4049	14	–	bmr4911	1	–	bmr6028	1	1	bmr7032	3	–
bmr4054	4	–	bmr4912	3	–	bmr6029	3	–	bmr7035	1	–
bmr4055	10	–	bmr4918	9	–	bmr6031	1	1	bmr7036	1	–
bmr4060	8	–	bmr4924	3	–	bmr6032	1	–	bmr7057	8	8
bmr4069	1	–	bmr4929	4	–	bmr6044	3	–	bmr7067	3	–
bmr4070	2	–	bmr4931	4	–	bmr6046	13	–	bmr7069	1	–
bmr4072	3	–	bmr4932	4	–	bmr6051	2	–	bmr7079	1	–
bmr4076	20	–	bmr4934	3	–	bmr6058	1	–	bmr7083	6	–
bmr4078	1	–	bmr4957	18	–	bmr6060	18	–	bmr7087	45	–
bmr4079	14	–	bmr4961	3	–	bmr6070	4	–	bmr7089	1	–
bmr4080	39	–	bmr4964	6	–	bmr6071	16	–	bmr7092	6	–
bmr4081	6	–	bmr4965	8	–	bmr6072	6	–	bmr7094	1	1
bmr4082	6	6	bmr4968	11	–	bmr6080	9	–	bmr7103	26	–
bmr4083	8	–	bmr4969	19	–	bmr6082	20	–	bmr7104	4	4
bmr4084	8	–	bmr4971	2	–	bmr6089	7	–	bmr7106	2	–
bmr4085	6	–	bmr4972	27	–	bmr6091	5	–	bmr7108	14	–
bmr4087	10	–	bmr4978	12	–	bmr6092	11	–	bmr7109	10	10
bmr4094	2	–	bmr4981	2	2	bmr6096	18	–	bmr7112	9	–
bmr4095	23	23	bmr4984	9	–	bmr6104	38	–	bmr7115	3	–
bmr4098	9	–	bmr4985	12	–	bmr6110	3	–	bmr7129	43	–
bmr4110	3	–	bmr4991	9	–	bmr6121	29	–	bmr7136	2	–
bmr4114	2	–	bmr5000	1	–	bmr6122	3	3	bmr7142	5	–
bmr4117	3	–	bmr5006	6	–	bmr6123	2	–	bmr7159	2	–
bmr4121	10	10	bmr5022	1	–	bmr6127	17	–	bmr7160	2	–
bmr4127	28	–	bmr5024	4	–	bmr6132	42	–	bmr7172	1	–
bmr4131	23	–	bmr5028	2	–	bmr6133	17	–	bmr7174	1	–
bmr4132	15	–	bmr5030	2	–	bmr6135	8	–	bmr7184	18	–
bmr4133	1	–	bmr5033	2	–	bmr6137	12	–	bmr7201	5	–
bmr4136	2	–	bmr5039	1	–	bmr6150	19	–	bmr7210	4	–
bmr4143	10	–	bmr5051	1	–	bmr6159	9	–	bmr7211	3	–
bmr4145	2	–	bmr5058	6	–	bmr6160	10	–	bmr7220	10	–
bmr4149	1	–	bmr5059	5	–	bmr6176	1	–	bmr7241	1	–
bmr4156	6	–	bmr5060	11	–	bmr6177	9	–	bmr7251	1	–
bmr4157	11	–	bmr5067	2	–	bmr6178	5	–	bmr7259	1	–
bmr4158	8	–	bmr5069	1	–	bmr6185	12	–	bmr7264	7	–
bmr4159	1	–	bmr5070	10	–	bmr6188	2	2	bmr7310	1	–

continued

Protein	Static	MD	Protein	Static	MD	Protein	Static	MD	Protein	Static	MD
bmr4161	6	–	bmr5077	7	–	bmr6195	39	–	bmr7314	1	–
bmr4162	2	–	bmr5082	5	–	bmr6198	5	–	bmr7315	1	–
bmr4163	1	–	bmr5083	17	–	bmr6199	1	–	bmr7324	2	–
bmr4168	1	–	bmr5093	5	–	bmr6202	1	–	bmr7357	1	–
bmr4173	3	–	bmr5097	2	–	bmr6209	2	2	bmr7371	2	2
bmr4177	3	–	bmr5099	16	–	bmr6210	1	–	bmr7386	10	–
bmr4184	5	–	bmr5104	1	–	bmr6221	–	1	bmr7399	1	–
bmr4193	7	–	bmr5126	1	–	bmr6222	1	–	bmr7400	3	–
bmr4195	1	–	bmr5128	1	–	bmr6225	4	–	bmr7423	1	–
bmr4197	5	–	bmr5129	12	–	bmr6231	7	–	bmr7424	1	–
bmr4202	5	–	bmr5143	1	–	bmr6233	1	–	bmr7425	1	–
bmr4204	5	–	bmr5149	1	–	bmr6238	4	–	bmr10004	32	–
bmr4206	29	–	bmr5173	3	–	bmr6241	1	–	bmr10015	2	–
bmr4207	5	–	bmr5175	1	–	bmr6252	1	–	bmr10027	1	–
bmr4210	25	–	bmr5176	1	–	bmr6254	1	1	bmr10043	1	–
bmr4211	4	–	bmr5182	1	–	bmr6255	12	–	bmr10064	2	–
bmr4216	6	–	bmr5194	16	–	bmr6256	1	–	bmr10065	2	–
bmr4217	3	–	bmr5207	1	–	bmr6257	1	–	bmr10118	1	1
bmr4219	2	–	bmr5210	2	–	bmr6264	2	–	bmr11062	2	–
bmr4224	4	–	bmr5217	10	–	bmr6268	17	–	bmr15001	23	–
bmr4236	3	–	bmr5218	1	–	bmr6271	1	–	bmr15002	1	–
bmr4237	3	3	bmr5232	1	–	bmr6279	16	–	bmr15014	12	–
bmr4241	1	–	bmr5242	2	–	bmr6282	1	–	bmr15037	4	–
bmr4242	8	–	bmr5257	1	–	bmr6285	2	–	bmr15074	7	–
bmr4254	2	2	bmr5261	11	–	bmr6292	2	–	bmr15085	1	–
bmr4255	4	–	bmr5262	1	–	bmr6298	1	–	bmr15129	1	–
bmr4256	33	–	bmr5264	9	–	bmr6299	18	–	bmr15156	3	–
bmr4257	5	–	bmr5267	5	–	bmr6300	2	–	bmr15161	1	–
bmr4261	3	–	bmr5275	1	–	bmr6313	1	–	bmr15166	6	–
bmr4278	4	–	bmr5284	1	–	bmr6315	1	–	bmr15204	1	–
bmr4280	1	–	bmr5285	1	–	bmr6317	28	–	bmr15232	2	–
bmr4290	1	–	bmr5311	2	2	bmr6321	1	–	bmr15242	1	–
bmr4295	2	–	bmr5312	4	–	bmr6322	1	–	bmr15320	1	–
bmr4297	1	–	bmr5313	8	–	bmr6334	3	–	bmr15332	1	–
bmr4299	30	–	bmr5317	19	–	bmr6338	11	–	bmr15333	1	–
bmr4300	29	–	bmr5320	14	–	bmr6339	22	–	bmr15355	1	–
bmr4301	1	–	bmr5322	1	–	bmr6342	6	–	bmr15372	6	–
bmr4305	6	–	bmr5326	17	–	bmr6344	3	–	bmr15394	1	–
bmr4307	17	–	bmr5328	1	1	bmr6349	1	–	bmr15399	3	–
bmr4308	11	–	bmr5329	3	3	bmr6354	10	–	bmr15406	15	–
bmr4309	1	–	bmr5342	3	–	bmr6355	6	6	bmr15412	2	–
bmr4318	13	–	bmr5347	36	–	bmr6356	3	–	bmr15420	32	–
bmr4327	4	–	bmr5354	18	–	bmr6362	12	–	bmr15421	7	–
bmr4333	3	–	bmr5356	7	–	bmr6371	3	–	bmr15424	15	–
bmr4334	4	–	bmr5359	4	–	bmr6375	11	11	bmr15429	45	–
bmr4335	10	–	bmr5364	1	–	bmr6376	2	2	bmr15498	17	–
bmr4336	20	–	bmr5375	1	1	bmr6380	1	1	bmr15503	4	–
bmr4340	26	–	bmr5393	11	–	bmr6400	2	–	bmr15507	2	–
bmr4343	1	–	bmr5398	20	–	bmr6401	1	–	bmr15508	1	–
bmr4348	1	1	bmr5402	33	–	bmr6406	10	–	bmr15521	2	–
bmr4350	1	–	bmr5408	1	–	bmr6407	1	–	bmr15548	1	–
bmr4360	3	–	bmr5410	1	–	bmr6410	7	–	bmr15560	4	–
bmr4370	20	–	bmr5454	3	–	bmr6442	3	–	bmr15564	1	–
bmr4378	20	–	bmr5455	9	–	bmr6447	2	–	bmr15594	1	–
bmr4380	1	–	bmr5457	2	–	bmr6448	2	–	bmr15669	1	–
bmr4393	10	–	bmr5459	3	–	bmr6452	1	–	bmr15692	1	–
bmr4394	11	–	bmr5464	1	–	bmr6455	12	–	bmr15756	2	2
bmr4396	1	–	bmr5473	24	–	bmr6458	10	–	bmr15757	1	–
bmr4401	2	–	bmr5475	1	–	bmr6460	23	–	bmr15783	3	–
bmr4417	2	–	bmr5485	10	–	bmr6465	1	–	bmr15785	10	–
bmr4421	8	8	bmr5492	3	–	bmr6466	4	4	bmr15792	1	–
bmr4422	1	–	bmr5495	1	–	bmr6469	1	1	bmr15800	1	–
bmr4425	1	–	bmr5499	1	1	bmr6476	2	–	bmr15826	2	–
bmr4426	1	–	bmr5512	1	–	bmr6484	3	–	bmr15852	1	–
bmr4427	5	–	bmr5516	7	–	bmr6493	13	–	bmr15866	1	–
bmr4428	1	–	bmr5540	12	–	bmr6498	9	–	bmr15928	37	–
bmr4429	1	–	bmr5555	38	–	bmr6500	1	–	bmr15952	1	–
bmr4431	14	–	bmr5563	4	4	bmr6501	5	–	bmr16024	1	–
bmr4437	29	–	bmr5565	6	–	bmr6503	4	–	bmr16074	1	–
bmr4443	1	–	bmr5571	29	–	bmr6506	9	–	bmr16169	2	–
bmr4449	10	–	bmr5572	1	–	bmr6519	29	–	bmr16171	4	–
bmr4451	3	–	bmr5583	1	–	bmr6520	3	–	bmr16228	12	–
bmr4452	1	–	bmr5589	1	–	bmr6525	2	–	bmr16235	1	–
bmr4458	8	–	bmr5594	2	–	bmr6527	49	49	bmr16243	2	–
bmr4459	8	–	bmr5595	9	–	bmr6536	1	–	bmr16294	8	–
bmr4460	1	–	bmr5606	1	–	bmr6538	1	–	bmr16310	7	–
bmr4461	1	–	bmr5609	1	–	bmr6540	1	–	bmr16321	4	–
bmr4466	17	–	bmr5610	8	–	bmr6541	1	–	bmr16328	3	–
bmr4467	1	–	bmr5621	1	–	bmr6554	5	–	bmr16334	11	–
bmr4473	2	–	bmr5622	1	–	bmr6555	1	–	bmr16362	1	–
bmr4475	10	–	bmr5629	8	–	bmr6563	6	–	bmr16367	1	–
bmr4486	2	–	bmr5650	2	–	bmr6564	4	–	bmr16379	32	–
bmr4494	11	–	bmr5654	4	–	bmr6571	5	–	bmr16447	1	–

continued

Protein	Static	MD	Protein	Static	MD	Protein	Static	MD	Protein	Static	MD
bmr4516	4	–	bmr5662	1	–	bmr6575	3	–	bmr16466	20	–
bmr4527	1	–	bmr5665	6	–	bmr6578	9	–	bmr16493	2	–
bmr4553	33	–	bmr5668	21	–	bmr6579	6	–	bmr16494	2	–
bmr4558	21	–	bmr5670	2	–	bmr6580	6	–	bmr16502	2	–
bmr4567	1	–	bmr5677	10	–	bmr6583	1	–	bmr16523	16	–
bmr4568	1	–	bmr5689	15	15	bmr6585	27	–	bmr16524	34	–
bmr4579	3	–	bmr5696	1	–	bmr6587	1	–	bmr16525	34	–
bmr4581	3	3	bmr5697	1	–	bmr6588	1	–	bmr16526	33	–
bmr4587	8	–	bmr5701	12	–	bmr6589	1	–	bmr16527	32	–
bmr4590	20	20	bmr5706	1	–	bmr6590	3	–	bmr16528	34	–
bmr4591	1	–	bmr5710	11	–	bmr6592	1	–	bmr16541	1	–
bmr4615	4	–	bmr5729	14	–	bmr6596	1	–	bmr16587	1	–
bmr4619	2	–	bmr5740	19	–	bmr6597	31	–	bmr16616	5	–
bmr4628	1	–	bmr5741	36	–	bmr6598	1	–	bmr16668	1	–
bmr4637	20	–	bmr5744	1	–	bmr6600	10	–	bmr16670	2	–
bmr4638	1	–	bmr5749	2	–	bmr6612	2	–	bmr16722	1	–
bmr4639	1	–	bmr5754	1	–	bmr6621	4	–	bmr16723	1	–
bmr4642	2	–	bmr5767	2	–	bmr6629	1	–	bmr16847	1	–
bmr4644	2	–	bmr5778	1	–	bmr6635	1	–	bmr16878	14	–
bmr4661	1	–	bmr5779	2	–	bmr6639	3	–	bmr16904	13	–
bmr4666	3	–	bmr5783	1	–	bmr6647	4	–	bmr16925	2	–
bmr4674	41	–	bmr5789	4	–	bmr6648	1	–	bmr17008	3	–
bmr4677	3	–	bmr5798	10	10	bmr6649	1	1	bmr17357	4	–
bmr4684	13	–	bmr5801	2	2	bmr6654	9	–	bmr17393	23	–
bmr4686	4	–	bmr5806	1	–	bmr6688	49	–	bmr17415	1	–
bmr4688	17	–	bmr5808	1	–	bmr6717	8	–	bmr17476	1	–
bmr4696	4	–	bmr5810	1	–	bmr6718	7	7			
bmr4703	15	–	bmr5817	8	–	bmr6724	1	–			

Table 6–Exclusions: Proteins with omitted atom assignments in the static and molecular-dynamics populations.